Fine structure of a Gene

Microbiology VI

Introduction

- A **gene** is a specific sequence of **DNA** containing genetic information required to make a **specific protein**.

- Prokaryotic gene is **uninterrupted**.

- In **Eukaryotic gene** the coding sequences (**exons**) are separated by non-coding sequences called **introns**.

- In complex eukaryotes, introns account for more than 10 times as much DNA as exons.

History of genes:



- The classical principles of genetics were deduced by **Gregor Mendel** in 1865 on the basis of breeding experiments with peas.

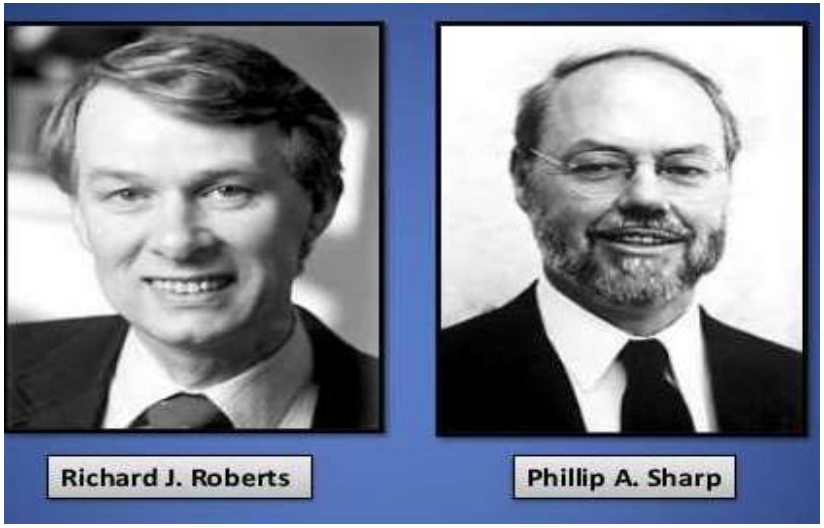
- He assumed that each trait is determined by a pair of **inherited 'factors'** which are now called **gene**.

- Mendel's work was rediscovered in 1900 by Hugo de Vries, Correns and Tschermak.

- Wilhelm Johannsen coined the term '**GENE**' in 1909.
- William Bateson in 1905 coined the term **genetics.**

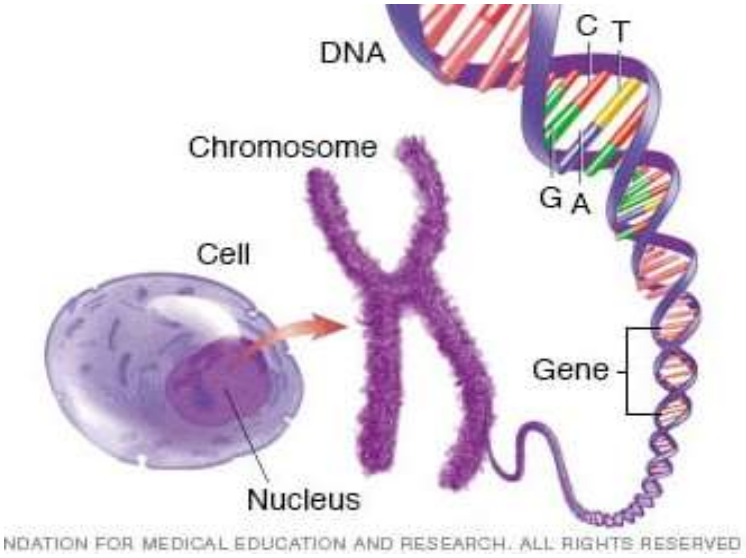
- In 1972, Walter Fiers and his team determined the sequence of a gene in **bacteriophage** MS2 coat protein.

- Richard J Roberts and Phillip Sharp found the gene can be split into **segments** making it possible that a **single gene** might be coding for several **proteins**.



Gene:

- The gene is the functional unit of **heredity**.
- Each gene is a segment of DNA that give rise to a **protein** product or RNA.
- A gene may exist in alternative forms called **alleles**.
- Chromosome in fact carry genes.
- Each chromosome consists of a linear array of genes



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Genes are classified based on their activity as:

1. House keeping genes:

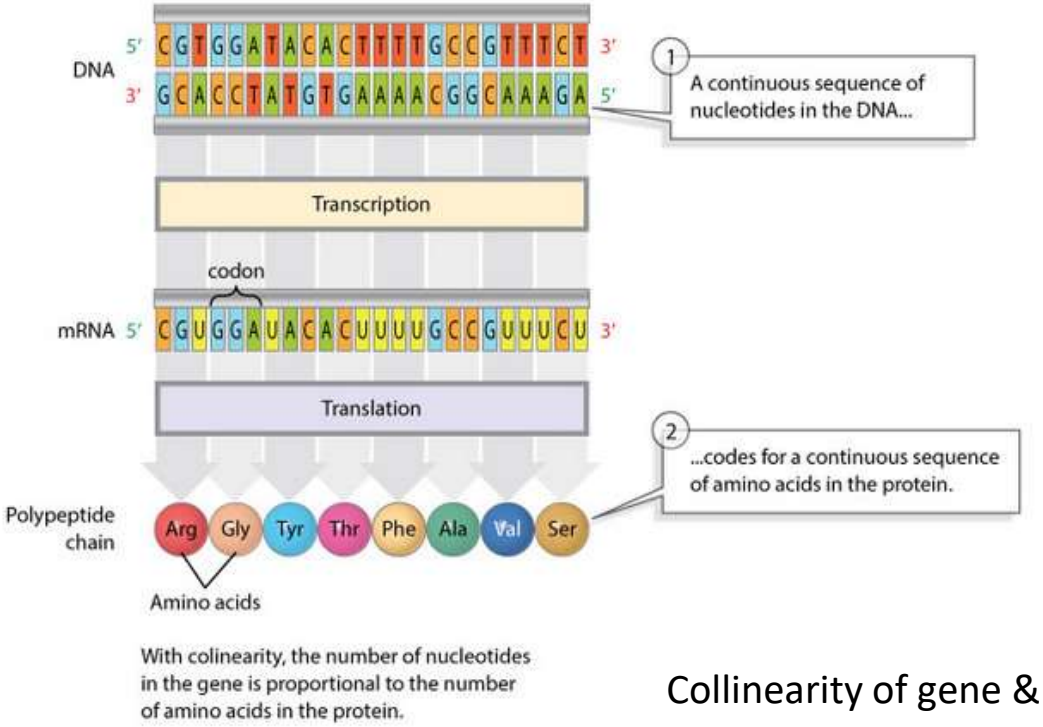
- are constitutive genes that are required for the maintenance of basic cellular functions
- They're expressed in all cells of an organism under normal conditions.

2. Specific genes: expresses for a specific type of protein when required by the cell.

Prokaryotic Gene structure

Structural features:

- Simple gene structure.
- Small genomes (0.5 to 10 million bp).
- Prokaryotic genes are collinear with their proteins.



Collinearity of gene & protein

Genes have three regions:

- Coding region
- Promoter elements
- Terminal region or terminator

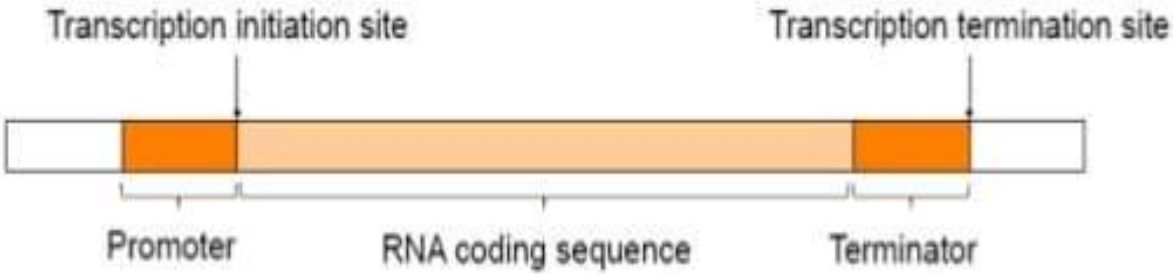
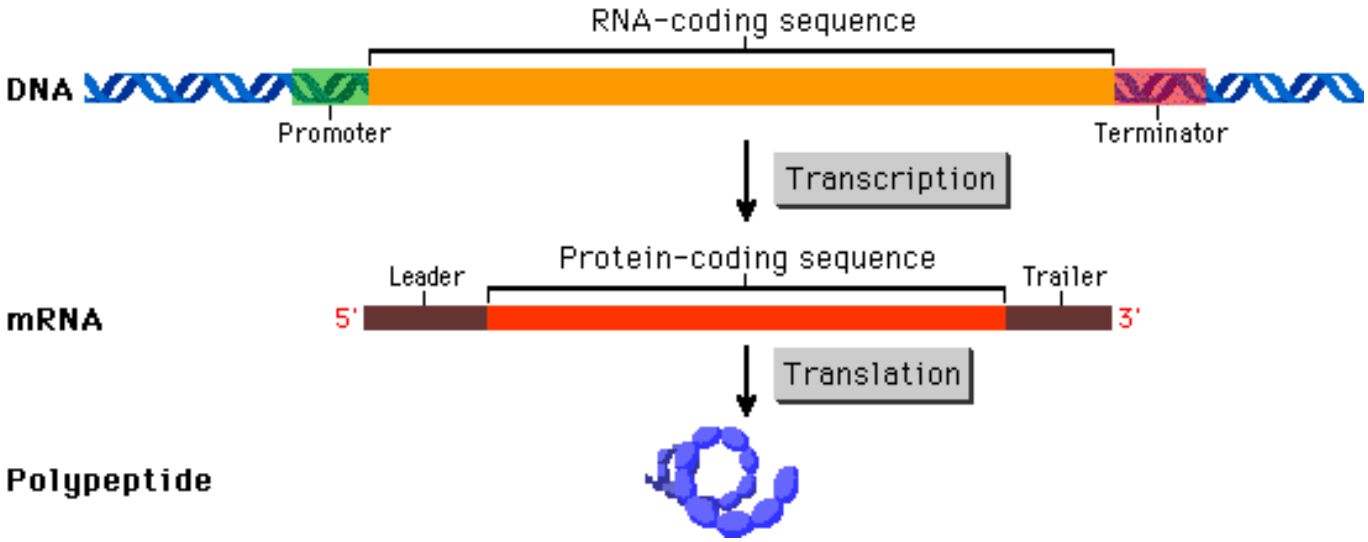


Figure 2: Prokaryotic gene structure

Coding region:

- The DNA sequence that is copied into **RNA molecule** (RNA transcript)
- Starts with an **initiator** codon and ends with **termination** codon.
- No introns (uninterrupted).
- Collinear to its mRNA.



Promoter elements:

- The upstream elements from the start of the coding region include **promoter** sequences.

- 50 to 100 nucleotides upstream of the start codon transcriptional initiation site or START site.

- Any nucleotide present on the **left** is denoted by **(-)** symbol and the region is called upstream element. Ex: - 10, -20, -35 etc.

- Start site symbolized by **+1**.
- Any sequence to the **right** of the start is downstream elements and numbered as +10,+35 etc.

Promoter elements in genes:

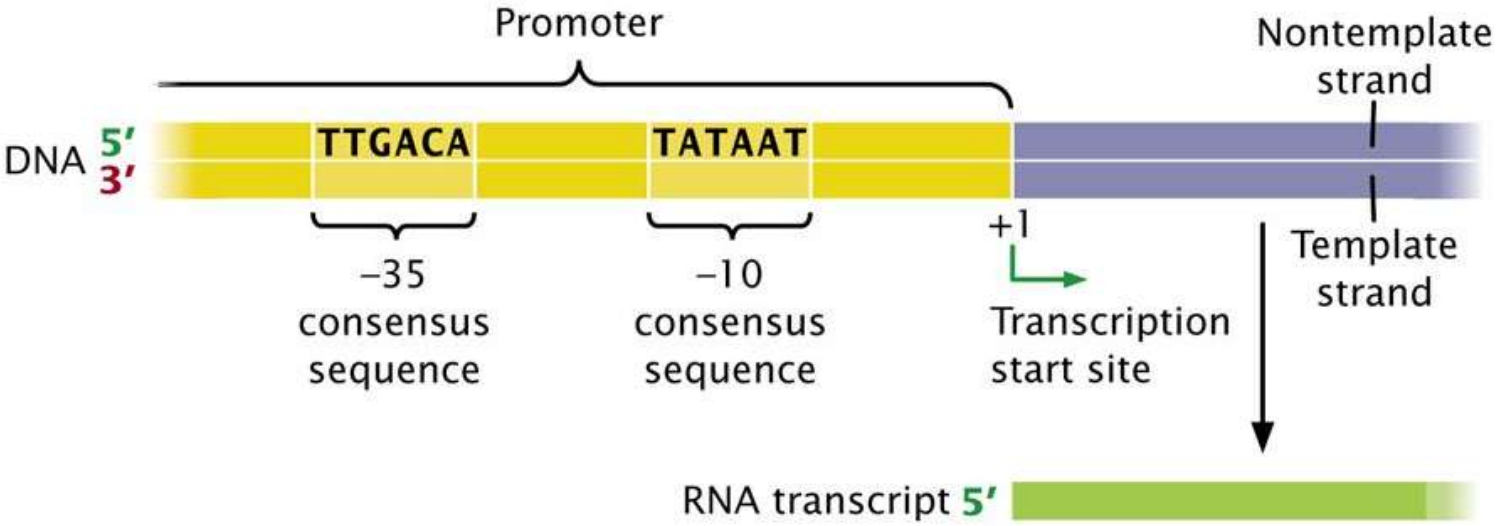
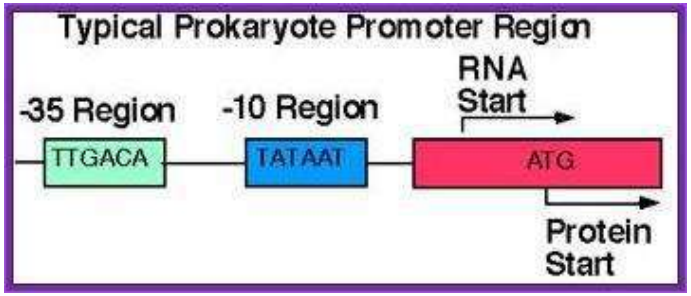
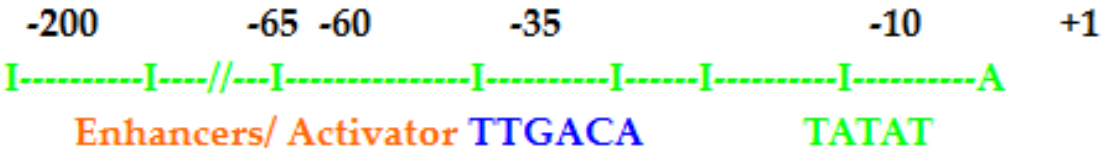


Fig_13-11 *Genetics, Second Edition* © 2005 W.H. Freeman and Company

- At **-10** there is a sequence **TATAAT** or **PRIBNOW BOX**.
- At **-35** another consensus sequence TTGACA
- These two are the most important **promoter elements**.
- Recognized by transcription factors.



- At -65 to -60 activator elements. Activation of the polymerase.
- At -200 to -1000 enhancer sequence. Enhances transcription by 100 to 200 folds.



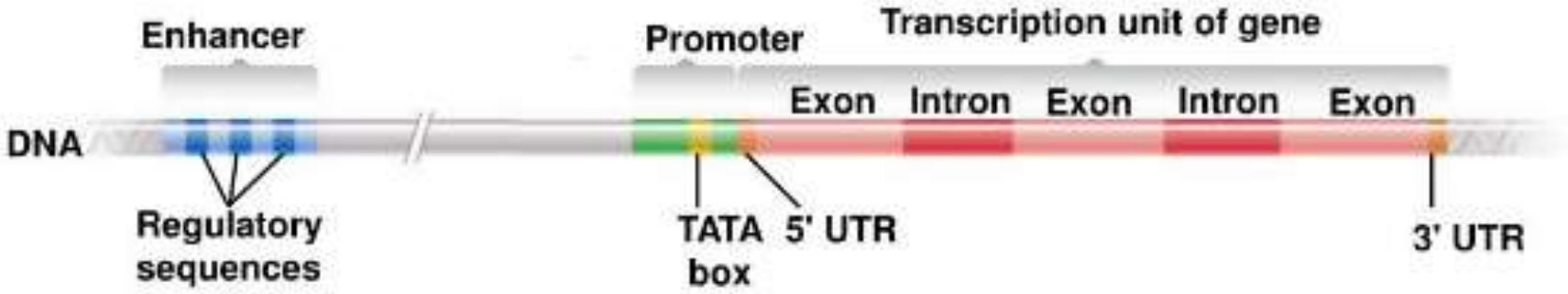
Terminal region of the gene:

- Sequences for the termination of transcription.
- It takes place by **Rho** dependent mode or Rho independent mode.

Eukaryotic gene structure

- Exons
- Introns
- Promoter sequences
- Terminator sequences
- Upstream sequences
- Downstream sequences
- Enhancers and silencers (upstream or downstream)
- **Signals:**
  - Upstream sequence signal for **addition of cap**.
  - Downstream sequences signal for addition of **poly A tail**.

Eukaryotic gene structure



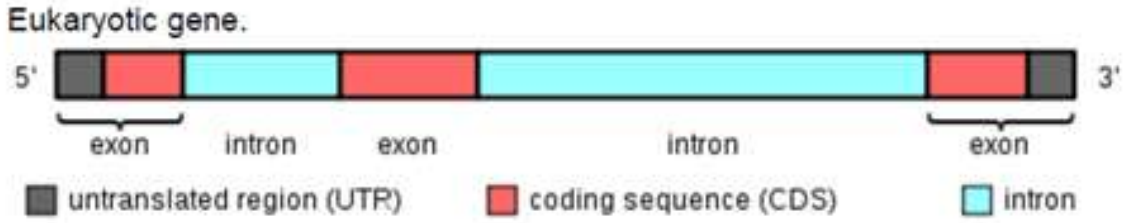
Exons and Introns

EXONS:

- **Coding sequence** that are transcribed and translated.
- Code for **amino acids** in the polypeptide chain.
- Vary in number, sequence and length.
- A gene starts and ends with exons. (5’ to 3’).
- Some exon includes untranslated (**UTR**) region.

INTRONS:

- **Noncoding** sequences that **separate** the coding sequences.
- Any nucleotide sequence that are **removed** when the primary transcript is processed to give the mature **RNA** are called introns.
- All introns share the base sequence **GT** in the 5’end and AG in the 3’end.
- Introns were 1st discovered in 1977 independently by Phillip Sharp and Richard Roberts.



PROMOTERS:

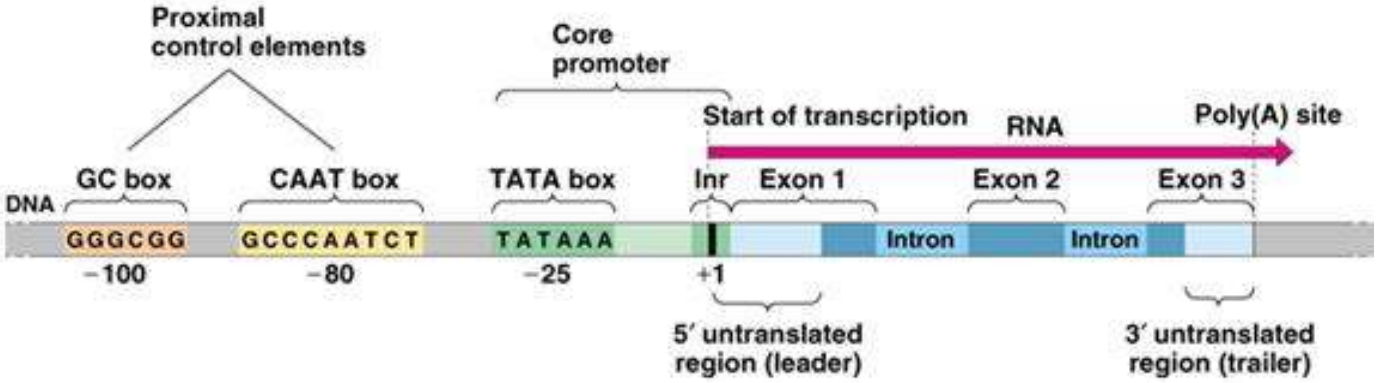
- A promoter is a regulatory region of DNA located upstream controlling gene expression.

1. **Core promoter:** transcription start site (-34)

- Binding site for RNA polymerase.
- General transcription factor binding sites.

2. **Proximal promoter:** contain primary regulatory element

- Approx. -250, specific transcription factor binding sites.
- **TATA** box or **hogness box** (-25 to -80) and **CAAT** (upstream TATA) are two distinct sequences.
- These together are responsible for binding of **RNA polymerase II** which is responsible for **transcription.**



UPSTREAM (5’end):

- 5’UTR serve several functions including **mRNA transport** and **initiation** of translation.
- Signal for addition of **cap** (7 methyl guanisine) to the 5’end of the mRNA.
- The cap facilitates the initiation of translation.
- **Stabilization** of mRNA.

DOWNSTREAM (3’end):

- 3’UTR serves to add mRNA stability and **attachment** site for poly-A-tail.
- The translation termination codon TAA.
- AATAA sequence signal for addition of **poly A tail**.

TERMINATOR:

- Recognized by RNA polymerase as a signal to **stop** transcription.

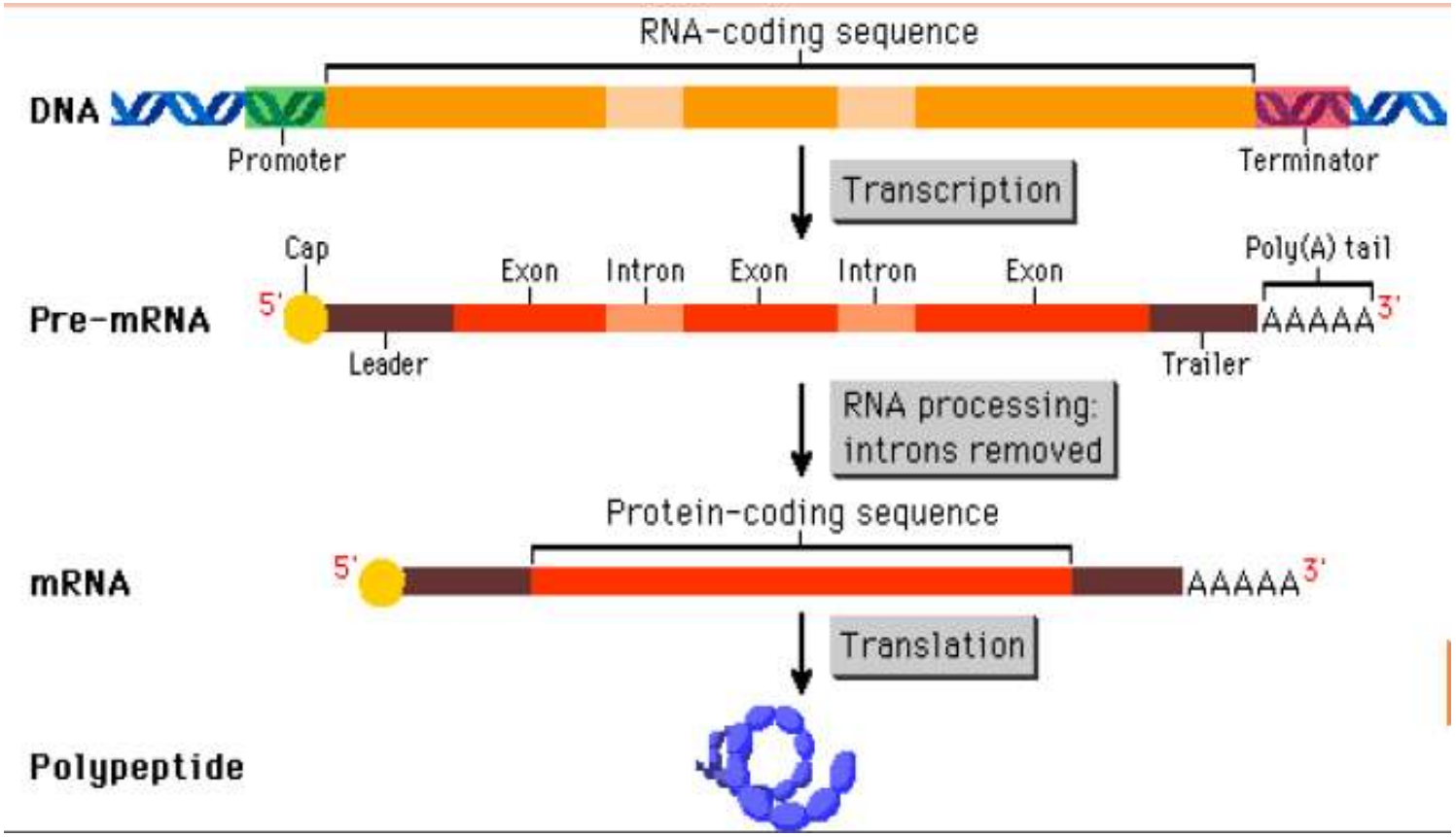
ENHANCER:

- Enhances the **transcription** of a gene.
- Upto few thousand bp upstream.

SILENCERS:

- **Reduce** or shut off the expression of a near by gene.

Eukaryotic gene structure & function



Significance of introns:

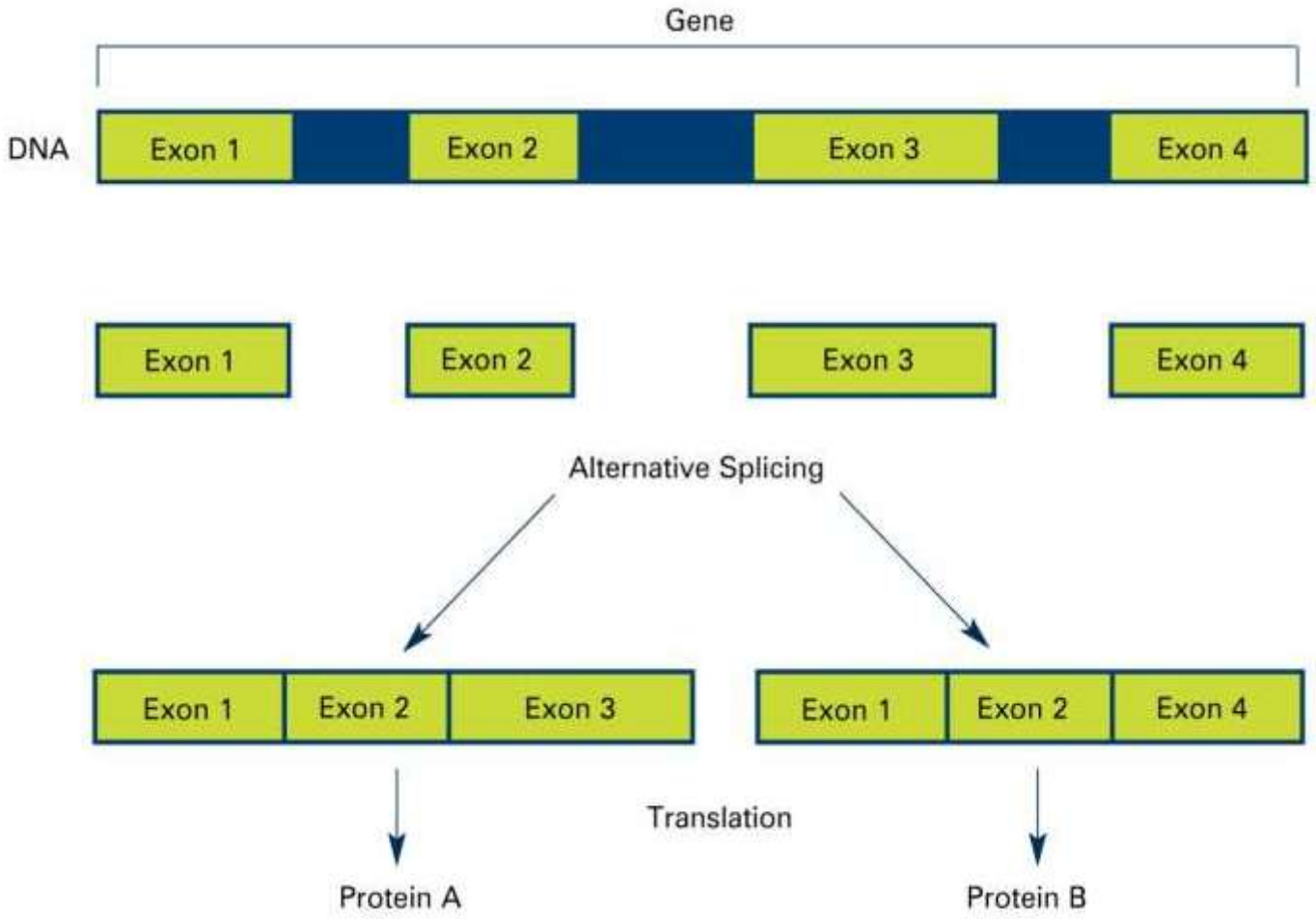
- Introns don't specify the synthesis of proteins but have other important cellular activities.

- Many introns encodes **RNA's** that are major **regulators** of gene expression.

- Contain regulatory sequences that **control** trancription and **mRNA processing**.

- Introns allow **exons** to be **joined** in different combinations (**alternative splicing**), resulting in the synthesis of **different proteins** from the **same gene**.

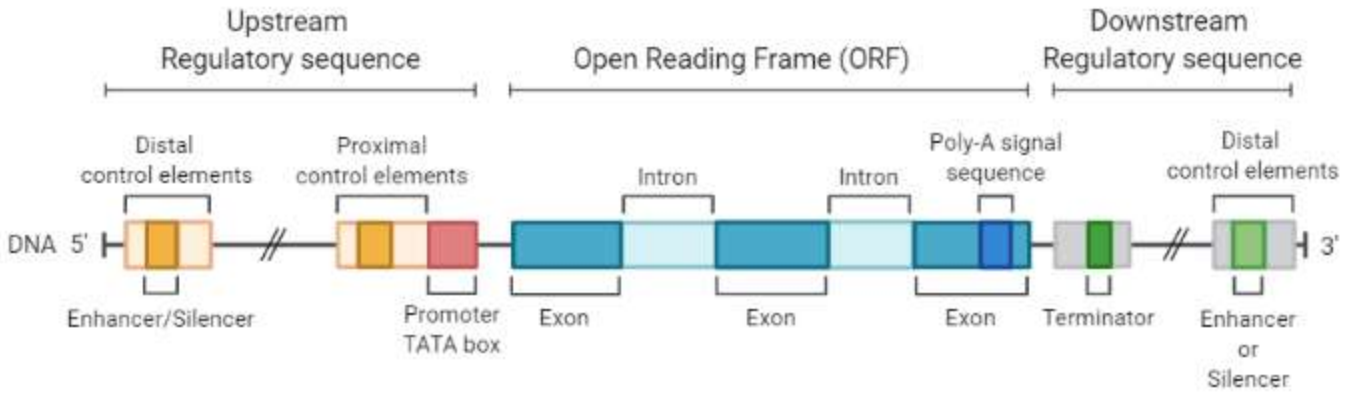
- Important role in **evolution** by facilitating recombination between **exons** of different genes (**exon shuffling**).



Intron significance: alternative splicing

Comparison of eukaryotic & prokaryotic gene structure

Eukaryotic Gene Structure



Prokaryotic Gene Structure

